

Day: _____

2022

Date: _____

SC-241

Q#1 → Short Answers!

i → Find the domain of
 $f(x, y) = \sqrt{xy}$.Solution: here $f(x, y) = \sqrt{xy}$

$$xy \geq 0$$

Divide each term in $xy \geq 0$ by

'y' } simplify:

$$\frac{xy}{y} \geq \frac{0}{y}$$

$$\cancel{y} \quad y$$

$$x \geq \frac{0}{y}$$

$$x \geq 0$$

The domain is: all values of x that
make the expression defined interval
notation.

$$[0, \infty)$$

set builder Notation: $\{x | x \geq 0\}$

Day: _____

Date: _____

ii- Find iterated integral

$$\int_1^3 \int_1^5 \frac{\ln y}{xy} dy dx$$

Solution

$$\int_1^3 \int_1^5 \frac{\ln y}{xy} dy dx$$

$$\int_1^3 \frac{1}{x} \int_1^5 \frac{\ln y}{y} dy dx$$

$$\int_1^3 \frac{1}{x} \int_1^5 \ln y \cdot \frac{1}{y} dy dx$$

$$\int_1^3 \frac{1}{x} \left[\frac{\ln y^2}{2} \right]_1^5 dx$$

$$\int_1^3 \frac{1}{x} \left(\frac{(\ln 5)^2}{2} - \frac{\ln(1)}{2} \right) dx$$

$$\frac{1}{2} \int_1^3 \frac{1}{x} \left((\ln 5)^2 - \ln 1 \right) dx$$

$$\frac{1}{2} \ln x \Big|_1^3 \left(\frac{(\ln 5)^2}{2} \right) \quad \because \ln 1 = 0$$

$$\frac{1}{2} (\ln 3 - \ln 1) \left(\frac{(\ln 5)^2}{2} \right)$$

$$\frac{1}{2} \ln 3 \left(\frac{(\ln 5)^2}{2} \right) \Rightarrow \ln 3 \left(\frac{(\ln 5)^2}{2} \right)$$

Day: _____

Date: _____

iii → Evaluate the integral

$$\int_C y \, dx + x \, dy + x \, dz, \quad 1 \leq t \leq 4$$

$$C: x = \sqrt{t}, \quad y = t, \quad z = t^2, \quad 1 \leq t \leq 4.$$

$$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$$

$$dx = \frac{1}{2\sqrt{t}} dt \quad dy = 1 dt \quad dz = 2t dt$$

$$dx = \frac{1}{2\sqrt{t}} dt \quad dy = 1 dt$$

$$= \int_C \frac{1}{2\sqrt{t}} dt + t^2 dt + \sqrt{t} \cdot 2t dt$$

$$= \int_C \frac{1}{2\sqrt{t}} + t^2 + 2t^{3/2} \cdot dt$$

$$= \frac{1}{2} \int_1^4 (t)^{1/2} dt + \int_1^4 t^2 dt + 2 \int_1^4 t^{3/2} dt$$

$$= \frac{1}{2} \frac{t^{3/2}}{3/2} + \frac{t^3}{3} + 2 \frac{t^{5/2}}{5/2}$$

$$= \frac{t^{3/2}}{3} + \frac{1}{3} t^3 + \frac{4}{5} t^{5/2}$$

$$= \left[\frac{1}{3} t^{3/2} + \frac{1}{3} t^3 + \frac{4}{5} t^{5/2} \right]_1^4$$

$$= \frac{1}{3} \left[(4)^{3/2} - (1)^{3/2} \right] + \frac{1}{3} \left[(4)^3 - (1)^3 \right]$$

$$+ \frac{4}{5} \left[(4)^{5/2} - (1)^{5/2} \right]$$

Day: _____

Date: _____

$$= \frac{1}{3} \left[(2)^{3/4} - 1 \right] + \frac{1}{3} \left[64 - 1 \right] + \frac{4}{5} \left[(2)^{5/4} - 1 \right]$$

$$= \frac{1}{3} [8-1] + \frac{1}{3} [63] + \frac{4}{5} [32-1]$$

$$= \frac{1}{3} [7] + \frac{1}{3} [63] + \frac{4}{5} (31)$$

$$= \frac{7}{3} + \frac{63}{3} + \frac{124}{5}$$

$$= \frac{35 + 315 + 372}{15}$$

$$= \frac{722}{15}$$

iv- Fourier sine series for
 $f(x) = 1$ in $(0, \pi)$.

$$f(x) = \sum_{n=1}^{\infty} [b_n \sin nx]$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx$$

$$= \frac{1}{\pi} \int_0^{\pi} 1 \sin n \, dn$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin n \, dn$$

Day: _____

Date: _____

$$= -\frac{1}{\pi} \left[\frac{\cos nx}{n} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\left(\frac{-1}{n} \right) - \left(\frac{-1}{n} \right) \right]$$

$$= \frac{1}{\pi} \left[1 - \left(\frac{-1}{n} \right)^n \right]$$

$$= \sum_{n=1}^{\infty} \left[\frac{1}{\pi} \left(\frac{1 - (-1)^n}{n} \right) \sin nx \right]$$

For $n=1$

$$\frac{1}{\pi} \left(\frac{1}{\pi} \left(1 - \left(\frac{-1}{2} \right) \right) \cdot \sin x \right)$$

$$\boxed{\frac{1}{\pi^2} 2 \sin x}$$

For $n=2$

$$\frac{1}{\pi} \left(\frac{1 - (-1)^2}{1} \sin 2x \right) = 0$$

$v \rightarrow$ find cosine transformation of

$$f(x) = e^{-x}$$

$$F_c(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx$$

$$F_c(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \cos bx \, dx$$

= ~~1~~ here

$$\int e^{ax} \cos bx \, dx =$$

$$\frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

$\rightarrow$ where $a = (-1)$

$$b = s$$

Day: _____

Date: _____

Now

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \cos sx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-x}}{(-1)^2 + s^2} \{ (-1) \cos sx + s \sin sx \} \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-x}}{1+s^2} (-\cos sx + s \sin sx) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-\infty}}{1+s^2} (-\cos \infty + s \sin \infty) - \right.$$

$$\left. \frac{e^0}{1+s^2} (-\cos 0 + s \sin 0) \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[0 - \frac{1}{1+s^2} (-1+0) \right]$$

$$= \sqrt{\frac{2}{\pi}} + \frac{1}{1+s^2}$$

$$\sqrt{\frac{2}{\pi}} \left(\frac{1}{1+s^2} \right) \frac{1}{2}$$

Day: _____

Date: _____

vi- determine even | odd

$$f(x) = |x^3 - x|$$

→ Even function:

$$f(-x) = f(x) \quad \forall x \text{ in domain}$$

→ Odd function:

$$f(-x) = -f(x) \quad \forall x \text{ in the domain}$$

$$f(-x) = ?$$

$$f(x) = |x^3 - x|$$

put $x = -x$

$$f(-x) = |(-x)^3 - (-x)|$$

$$f(-x) = |-x^3 + x|$$

$$f(-x) = |x^3 - x|$$

→ even function as $f(-x) = f(x)$

Note Absolute value function $|x|$ is even } cubic function x^3 is odd. However the function $|x^3 - x|$ is an even function.

Day: _____

Long Question

Date: _____

- ① Maximum + Minimum Value
of saddle point(s) of the function
 $f(x, y) = y^3 + 3x^2y - 6x^2 - 6y^2 + 2$

Partial diff w.r.t x

$$f_x = y^2 + 3x^2y - 6x^2 - 6y^2 + 0$$

$$f_x = 6xy - 12x \quad \text{--- ①}$$

Again p.d $\rightarrow$

$$f_{xx} = 6y - 12$$

Partial diff w.r.t y

$$f_y = 3y^2 + 3x^2 - 12y$$

$$f_{yy} = 6y - 12$$

$$6y - 12 = 0 \quad \text{--- ②}$$

$$3y^2 - 3x^2 + 12y = 0 \quad \text{--- ③}$$

$$6y = 12$$

$$\boxed{y = 2} \quad \rightarrow \text{put in ③}$$

$$3(2)^2 - 3x^2 + 12(2) = 0$$

$$12 - 3x^2 + 24 = 0$$

$$-3x^2 + 36 = 0$$

$$3x^2 = 36$$

$$x^2 = \frac{36}{3}$$

$$x^2 = 12$$

$$\boxed{x = \pm \sqrt{12}}$$

$$\boxed{x = \pm 12}$$

Day: _____

Date: _____

put value of $y = 2$

$$f_{xx} = 6(2) - 12$$

$$(12 - 12)$$

$$\boxed{f_{xx} = 0}$$

$$f_{yy} = 6y - 12$$

$$6(2) - 12$$

$$12 - 12 = 0$$

$$\boxed{f_{yy} = 0}$$

$$f_{xy} = 6\sqrt{12}$$

$$D = AC - B^2$$

$$A = f_{xx} \quad (0)(0) - (6\sqrt{12})^2$$

$$B^2 = f_{xy}^2 \quad = 36$$

$$C = f_{yy}$$

$D < 0$ the P is saddle point

② Double Integral

$\iint_D x \cos y \, dA$, D is bounded by

$$y = 0, y = x^2, x = 1$$

$$\begin{array}{l|l} y \rightarrow 0 - x^2 & y = x^2 \\ x \rightarrow 0 \rightarrow 1 & 0 = x^2 \\ & x = 0 \end{array}$$

Day: _____

Date: _____

$$\int_0^1 \int_0^{x^2} x \cos y \, dy \, dx$$

$$= \int_0^1 \int_0^{x^2} x \cos y \, dy \, dx$$

$$= \int_0^1 x \sin y \Big|_0^{x^2} dx$$

∵ $\sin 0 = 0$

$$= \int_0^1 x [\sin(x^2) - \sin(0)] dx$$

$$= \int_0^1 x [\sin x^2] dx$$

$$= \int_0^1 x \sin x^2 dx$$

$$= \frac{1}{2} \int_0^1 \sin x^2 \cdot 2x \, dx$$

$$= \frac{1}{2} \int_0^1 \sin x^2 \, dx$$

$$= \frac{1}{2} \left[-\cos x^2 \Big|_0^1 \right]$$

$$= -\frac{1}{2} \left[\cos x^2 \Big|_0^1 \right]$$

$$= -\frac{1}{2} [\cos(1) - \cos(0)]$$

$$= -\frac{1}{2} \cos 1 - 1$$

$$= \sqrt{\frac{-\cos 1 - 1}{2}} \quad \text{Ans}$$

Day: _____

Date: _____

③ line integral $\rightarrow$

$$\int_R \vec{F} \cdot d\vec{r}, \vec{F}(x, y, z) =$$

$$(x+y)\hat{i} + (y-z)\hat{j} + z^2\hat{k},$$

C is given by the vector function

$$\vec{r}(t) = t^2\hat{i} + t^3\hat{j} + t^2\hat{k},$$

$$0 \leq t \leq 1$$

find dr

$$\frac{dr}{dt} = \frac{d}{dt}(t^2)\hat{i} + \frac{d}{dt}(t^3)\hat{j} + \frac{d}{dt}(t^2)\hat{k}$$

$$dr = (2t\hat{i} + 3t^2\hat{j} + 2t\hat{k})dt$$

Since $r(t) = t^2, t^3, t^2$ substitute into F

$$(t^2 + t^3)(2t)$$

$$(t^2 + t^3)\hat{i} + (t^3 - t^2)\hat{j} + (t^4)\hat{k}$$

$$F \cdot dr = (t^2 + t^3)(2t) + (t^3 - t^2)(3t^2) + (t^4)(2t)$$

$$= 2t^3 + 2t^4 + 3t^5 - 3t^4 + 2t^5$$

Day: _____

Date: _____

$$= \int_0^1 (2t^3 - t^4 + 5t^5) dt$$

$$= \left[\frac{2t^4}{4} \right]_0^1 - \left[\frac{t^5}{5} \right]_0^1 + 5 \left[\frac{t^6}{6} \right]_0^1$$

$$= \frac{2}{4} - \frac{1}{5} + \frac{5}{6}$$

$$= \frac{10 - 4}{20} + \frac{5}{6}$$

$$= \frac{6}{20} + \frac{5}{6}$$

$$= \frac{36 + 100}{120} = 7 \frac{136}{120} \quad \left| \frac{136}{120} \right|$$